

Segmentation Engine Documentation

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1 Scope

This document describes the Segmentation Engine, a C++ HTTP service for route analysis and segment extraction. It has been prepared in accordance with ISO/IEC/IEEE 26514:2008, which defines structure and content for software user documentation.

2 Normative References

- ISO/IEC/IEEE 26514:2008 – Systems and software engineering – Requirements for designers and developers of user documentation.
- ISO 8601:2019 – Date and time representation.

3 Terms and Definitions

Segment A contiguous portion of a route, represented as GeoJSON geometry and associated metadata.

Route signal A sampled representation of map variables (elevation, curvature, ...) indexed by distance along the route.

Bounding box Geographic region defined by west, south, east and north coordinates.

4 System Overview

The engine exposes a RESTful API that receives map data, builds route signals, performs wavelet based segmentation and persists results to a MySQL database. The major modules are:

- **core**: algorithmic components such as the route signal builder, segmentation logic and wavelet utilities.
- **http**: request routing and REST endpoint handlers built on `cpp-httplib`.
- **infra**: database adapter for MySQL persistence.
- **models**: C++ types representing OSRM responses, segments and configuration parameters.

5 Installation and Operation

5.1 Prerequisites

CMake 3.15 or newer, a C++17 capable compiler, Docker and a running MySQL server.

5.2 Build and Deployment

The provided script builds and runs the engine inside a Docker container:

```
./build_and_deploy.sh
```

It performs three steps:

1. Build an image tagged `segmentation_engine:latest`.
2. Remove any existing container named `segmentation_container`.
3. Launch a new container on port 5005 with MySQL parameters (`DB_HOST`, `DB_PORT`, `DB_USER`, `DB_PASS`, `DB_NAME`) supplied via the environment.

Internally, the script issues the following Docker commands:

```
docker build -t segmentation_engine:latest .
docker rm -f segmentation_container >/dev/null 2>&1 || true
docker run -d --name segmentation_container --network host \
  -e DB_HOST=127.0.0.1 -e DB_PORT=3306 -e DB_USER=routeseq_user \
  -e DB_PASS=changeme-user -e DB_NAME=routeseq -p 5005:5005 \
  segmentation_engine:latest
```

Container logs can be monitored via:

```
docker logs -f segmentation_container
```

For a manual build without Docker:

```
cmake -S . -B build
cmake --build build
./build/segmentation_engine
```

5.3 Configuration

Runtime settings, including the HTTP port and enabled endpoints, are stored in `config/settings.json`. Database credentials are compiled into `src/main.cpp` and should be configured prior to deployment.

5.4 Execution

After deployment the service is reachable on `http://localhost:5005` and mounts the `public` directory under `/static`. POST and GET routes are registered from the configuration file.

To consolidate endpoints and avoid Cross-Origin Resource Sharing (CORS) issues, the parent `segmentation` directory provides a lightweight Nginx proxy. From that directory run:

```
./proxy.sh
```

The script starts a container named `quick-proxy` that forwards port 8081 to the Segmentation Engine at 5005, exposing the service at `http://localhost:8081`. Additional proxy ports 8082 and 8083 target the viewer and database UI, allowing browser clients to interact with all components through a single origin.

6 REST API

Table 1 lists the available endpoints. JSON bodies use UTF-8 encoding and timestamps follow ISO 8601.

Table 1: REST endpoints

Method/Path	Description
POST /upload	Accepts a map JSON file and stores it under <code>uploads/</code> . Returns the storage path.
GET /view?map= <i>path</i>	Renders an interactive HTML viewer comparing raw OSRM geometry to the derived route signal.
GET /viewLab?map= <i>path</i>	Redirects to the Signal Lab user interface.
GET /lab/meta?map= <i>path</i>	Lists available uploads and optionally returns metadata for specific map.
POST /lab/re-sample	Builds a route signal and resamples requested variables onto a uniform distance grid.
POST /wavelet	Performs wavelet-based terrain segmentation and optionally persists segments to the database.
GET /segments	Retrieves previously persisted segments intersecting a geographic bounding box.

7 Algorithms

Segmentation is accomplished by constructing a uniformly sampled route signal from the uploaded map and applying a wavelet-based energy detector. High energy regions correspond to changes in elevation or curvature and are converted into segments. Utilities in `core/wavelets` implement the FFT based rolling energy calculation. Existing segments can be matched against new routes to avoid redundant processing.

8 Data Management

Segment metadata is stored in a MySQL schema defined by `schema.sql`. The `infra::MySQLSegmentDB` component manages connections, inserts new segments and queries by bounding box.

9 Error Handling and Logging

Fatal signals (segmentation faults, aborts, arithmetic errors) are trapped and a stack trace is printed to the server log. Endpoint handlers translate exceptions into HTTP 500 responses containing a diagnostic message.

10 Conformance

The Segmentation Engine and this document use ISO 8601 timestamps and the terminology defined in ISO/IEC/IEEE 26514:2008. All numeric values are expressed using SI units.